

Antichains among Divisor Sums of Divisors

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Abstract

For a positive integer n , let

$$S(n) = \{\sigma(d) : d \mid n\}$$

be ordered by divisibility, and let $a(n)$ be its width. We study how much of the divisor lattice of n survives under the map $d \mapsto \sigma(d)$. For every fixed prime-exponent pattern, the largest possible value of $a(n)$ is the width of a corresponding product of divisibility posets, and this maximum is attained infinitely often. In the squarefree case with $\omega(n) = r$, every width from 1 to $\binom{r}{\lfloor r/2 \rfloor}$ occurs, again infinitely often.

For repeated exponent patterns, Engel's limit theory reduces the asymptotic width problem to optimal representation variance. We show that for the divisibility poset $D_m = (\{1, \dots, m\}, |)$, the prime directions 2 and 3 already determine the exact variance. The structural reason is the critical dilation 4, lying strictly between 3 and the next prime 5. This yields an exact finite formula, the limit $V(D_m) \rightarrow 1.0367934201598\dots$, and asymptotics for repeated exponent patterns. The complementary minimum-width problem remains largely open.

Key words and phrases. divisibility poset; antichain; Sperner theory; sum-of-divisors function; optimal representation; divisor lattice.

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1 Introduction

For a positive integer n , consider

$$S(n) = \{\sigma(d) : d \mid n\}.$$

We order this finite set by divisibility and write

$$a(n) = \text{width}(S(n), |).$$

Thus $a(n)$ is the largest number of distinct divisor sums, taken from divisors of the same integer n , such that no one divides another.

For example, if $n = 6$,

$$S(6) = \{1, 3, 4, 12\}.$$

The two middle values are incomparable, so $a(6) = 2$. The initial values of the sequence are

$$1, 1, 1, 2, 1, 2, 1, 2, 2, 1, 1, 3, 1, 2, 2, 3, 1, 3, 1, 2, \dots$$



Figure 1: Two basic divisibility posets. The first nontrivial set $S(6) = \{1, 3, 4, 12\}$ is a diamond. For a prime fourth power, the divisor sums have the order structure of D_5 ; this is proved in Theorem 2.3.

Sets with no internal divisibility are usually called *primitive sets*; they form a classical meeting point of multiplicative number theory and extremal combinatorics [7, 10]. Here the map

$$d \mapsto \sigma(d)$$

sends the divisor lattice of one integer into the divisibility poset of the positive integers, and we ask how much order structure survives. For a prime power, divisibility among the values $\sigma(p^{k-1}) = 1 + p + \dots + p^{k-1}$ is controlled exactly by divisibility of the index k . With several prime factors, additional *cross-divisibilities* between different prime bases become the main arithmetic difficulty.

For a fixed exponent pattern $\mathbf{e} = (e_1, \dots, e_r)$, we show that the primes can be chosen so that only the index-forced relations remain. Hence the maximal problem becomes the width problem for

$$D_{e_1+1} \times \dots \times D_{e_r+1}.$$

In the squarefree case the extremal widths are 1 and the central binomial coefficient; a one-parameter threshold construction shows that every intermediate width occurs as well.

Repeated exponent patterns lead to powers of $D_m = (\{1, \dots, m\}, |)$, bringing in Engel's theory of optimal representations and limit Sperner theorems [4, 6]. Divisibility orders on initial intervals have also been studied through order dimension [9, 8]; here the relevant invariant is the minimum representation variance $V(D_m)$. We prove that the prime directions 2 and 3 already determine this variance. The critical dilation is 4, and

$$3 < 4 < 5$$

explains why 3 must be retained while every prime at least 5 is redundant. This yields an exact formula for $V(D_m)$, its limiting constant, and asymptotics for repeated exponent patterns.

Thus cross-divisibilities are eliminated in the maximal problem, controlled in the squarefree spectrum, and reduced to their active prime directions in the variance problem. The complementary minimum-width problem is discussed in Section 7 and remains largely open.

2 Basic properties and prime powers

Proposition 2.1. *If $m \mid n$, then $a(m) \leq a(n)$.*

Proof. Every divisor of m is a divisor of n , hence $S(m) \subseteq S(n)$. Every antichain in $S(m)$ is therefore an antichain in $S(n)$. $\square$

For a prime p , put

$$R_k(p) = 1 + p + \cdots + p^{k-1} = \frac{p^k - 1}{p - 1}.$$

Lemma 2.2. *For $i, j \geq 1$, one has $R_i(p) \mid R_j(p)$ if and only if $i \mid j$.*

Proof. If $i \mid j$, the divisibility follows from the factorization of $x^j - 1$. Conversely,

$$\gcd(R_i(p), R_j(p)) = R_{\gcd(i, j)}(p).$$

Thus $R_i(p) \mid R_j(p)$ forces $i = \gcd(i, j)$, hence $i \mid j$. $\square$

Theorem 2.3. *For every prime p and every $e \geq 0$,*

$$a(p^e) = \left\lfloor \frac{e+1}{2} \right\rfloor.$$

Proof. Since

$$S(p^e) = \{R_1(p), \dots, R_{e+1}(p)\},$$

Lemma 2.2 identifies this poset with D_{e+1} . The integers

$$\left\lfloor \frac{e+1}{2} \right\rfloor + 1, \dots, e+1$$

form an antichain. Conversely, partition $1, \dots, e+1$ according to odd part; every class is a chain of the form $u, 2u, 4u, \dots$. There are $\lceil (e+1)/2 \rceil$ such chains, so Dilworth's theorem [2] gives the result. $\square$

Corollary 2.4. *Every positive integer occurs as a value of $a(n)$.*

Proof. For $k \geq 1$, Theorem 2.3 gives $a(p^{2k-2}) = k$. $\square$

Proposition 2.5. *If $a(n) = 1$, then n is squarefree.*

Proof. If $p^2 \mid n$, then $p+1$ and p^2+p+1 belong to $S(n)$. The latter is larger than $p+1$ and is congruent to 1 modulo $p+1$, so these two values are incomparable. $\square$

Proposition 2.6. *Let $p < q$ be primes. Then*

$$a(pq) = \begin{cases} 1, & p+1 \mid q+1, \\ 2, & p+1 \nmid q+1. \end{cases}$$

Proof. We have

$$S(pq) = \{1, p+1, q+1, (p+1)(q+1)\}.$$

The only possible incomparability is between $p+1$ and $q+1$. $\square$

3 Squarefree integers

Suppose

$$n = p_1 \cdots p_r$$

is squarefree. For $I \subseteq [r]$,

$$\sigma\left(\prod_{i \in I} p_i\right) = \prod_{i \in I} (p_i + 1).$$

Theorem 3.1. *If n is squarefree and $\omega(n) = r$, then*

$$a(n) \leq \binom{r}{\lfloor r/2 \rfloor}.$$

Proof. Choose one subset $I \subseteq [r]$ representing each value in an antichain of $S(n)$. If $I \subseteq J$, then the product indexed by I divides the product indexed by J , which is impossible for two distinct values in the chosen antichain. The chosen subsets therefore form an antichain in the Boolean lattice B_r . Sperner's theorem [11] gives the bound. $\square$

Theorem 3.2. *For every $r \geq 1$, infinitely many squarefree integers n satisfy*

$$\omega(n) = r, \quad a(n) = 1.$$

Proof. The assertion is clear for $r = 1$. Proceed by induction. Suppose n is squarefree and $S(n)$ is a chain, and put $M = \sigma(n)$. By Dirichlet's theorem [3], there are infinitely many primes

$$q \equiv -1 \pmod{M}.$$

Choose one with $q \nmid n$. Then

$$S(nq) = S(n) \cup (q+1)S(n).$$

The two copies are chains, and every element of the first copy divides every element of the second because every element of $S(n)$ divides M and hence divides $q+1$. Thus the union is again a chain. $\square$

4 Fixed exponent patterns

The prime-power case suggests a natural model for a general exponent vector. If

$$n = p_1^{e_1} \cdots p_r^{e_r},$$

then every divisor sum is a product of repunits $R_k(p_i)$. If one ignores divisibilities between different prime bases, the index set is the product poset

$$P_{\mathbf{e}} = D_{e_1+1} \times \cdots \times D_{e_r+1}, \quad \mathbf{e} = (e_1, \dots, e_r).$$

The next theorem shows that this naive model is the exact maximal one: all additional cross-divisibilities can be avoided simultaneously.

Theorem 4.1. *For every fixed exponent vector $\mathbf{e}$,*

$$\boxed{\max_{p_1, \dots, p_r} a(p_1^{e_1} \cdots p_r^{e_r}) = \text{width}(P_{\mathbf{e}}),}$$

where the maximum is over distinct primes. Moreover, the maximum is attained for infinitely many choices of $p_1, \dots, p_r$.

Proof. For

$$\mathbf{k} = (k_1, \dots, k_r) \in P_{\mathbf{e}}$$

define

$$F(\mathbf{k}) = \prod_{i=1}^r R_{k_i}(p_i).$$

If $k_i \mid \ell_i$ for every i , then $F(\mathbf{k}) \mid F(\ell)$. Given an antichain among the distinct values $F(\mathbf{k})$, choose one index vector representing each value. No two chosen vectors can be comparable in $P_{\mathbf{e}}$, so they form an antichain. Therefore

$$a(p_1^{e_1} \cdots p_r^{e_r}) \leq \text{width}(P_{\mathbf{e}}).$$

It remains to realize exactly the coordinatewise divisibility relations. Put

$$m = \max_i (e_i + 1).$$

For every pair (i, k) with $2 \leq k \leq e_i + 1$, choose a distinct prime $q_{i,k} > m$ satisfying

$$q_{i,k} \equiv 1 \pmod{k},$$

and choose $g_{i,k}$ of exact multiplicative order k modulo $q_{i,k}$. For each fixed i , prescribe

$$p_i \equiv g_{i,k} \pmod{q_{i,k}}$$

for the primes belonging to the i -th coordinate, and

$$p_i \equiv 1 \pmod{q_{j,k}} \quad (j \neq i)$$

for all primes belonging to the other coordinates. The Chinese remainder theorem gives a reduced residue class, and Dirichlet's theorem supplies infinitely many primes in it. The p_i may therefore be chosen distinct.

For $2 \leq k \leq e_i + 1$ and $1 \leq u \leq e_i + 1$,

$$q_{i,k} \mid R_u(p_i) \iff k \mid u,$$

because p_i has exact order k modulo $q_{i,k}$. On the other hand, if $j \neq i$, then

$$R_u(p_j) \equiv u \pmod{q_{i,k}},$$

and $1 \leq u \leq m < q_{i,k}$, so $q_{i,k}$ divides none of the relevant factors from another coordinate.

Thus $q_{i,k}$ is a private witness for divisibility by k in the i -th coordinate. If some k_i does not divide ℓ_i , then q_{i,k_i} divides $F(\mathbf{k})$ but not $F(\ell)$. Consequently

$$F(\mathbf{k}) \mid F(\ell) \iff k_i \mid \ell_i \text{ for all } i.$$

Hence the divisibility poset is isomorphic to $P_{\mathbf{e}}$, proving equality. Since each prescribed residue class contains infinitely many primes, infinitely many choices attain the maximum. $\square$

Corollary 4.2. *For every $r \geq 1$,*

$$\boxed{\max_{\substack{n \text{ squarefree} \\ \omega(n)=r}} a(n) = \binom{r}{\lfloor r/2 \rfloor}.}$$

The maximum is attained infinitely often.

Proof. For the squarefree pattern, $P_e = D_2^r$, which is the Boolean lattice B_r . By Sperner's theorem,

$$\text{width}(B_r) = \binom{r}{\lfloor r/2 \rfloor},$$

and Theorem 4.1 gives the result, including infinitely many attaining choices. $\square$

Combining Theorem 3.2 and Corollary 4.2 shows that both squarefree extremal values are sharp and occur infinitely often.

5 The complete squarefree spectrum

For

$$\mathcal{W}_r = \{a(n) : n \text{ squarefree}, \omega(n) = r\}, \quad b_r = \binom{r}{\lfloor r/2 \rfloor},$$

we prove $\mathcal{W}_r = \{1, \dots, b_r\}$. The two ingredients are a one-prime extension that preserves width and a weighted construction for the new values between b_r and b_{r+1} .

Lemma 5.1 (Controlled one-prime extension). *Let n be squarefree, put $N = \sigma(n)$, and let $M \mid N$. Assume that $2 \mid M$ whenever $2 \mid N$. Then there are infinitely many primes $q > N$ for which*

$$q + 1 = Mk, \quad \gcd(k, N) = 1.$$

For every such q and all $x, z \in S(n)$,

$$\boxed{x \mid (q + 1)z \iff \frac{x}{\gcd(x, z)} \mid M.} \tag{5.1}$$

Proof. We first choose a residue class c modulo N . For every prime $\ell \mid N$ with $\ell \mid M$, require $c \not\equiv 0 \pmod{\ell}$. For every prime $\ell \mid N$ with $\ell \nmid M$, the prime ℓ is odd by the hypothesis on the factor 2; choose c modulo ℓ avoiding both 0 and M^{-1} . These local conditions can be lifted to the prime-power factors of N , and the Chinese remainder theorem gives c such that

$$\gcd(c, N) = 1, \quad \gcd(Mc - 1, N) = 1.$$

Since also $Mc - 1 \equiv -1 \pmod{M}$, we have

$$\gcd(Mc - 1, MN) = 1.$$

Dirichlet's theorem therefore gives infinitely many primes

$$q \equiv Mc - 1 \pmod{MN}.$$

Choosing one with $q > N$, write

$$q + 1 = M(c + Nt) = Mk.$$

Then $\gcd(k, N) = 1$.

Because n is squarefree, every element of $S(n)$ divides N . Put

$$g = \gcd(x, z), \quad x = gx', \quad z = gz',$$

so that $\gcd(x', z') = 1$. Then

$$x \mid (q+1)z \iff x' \mid Mxz'.$$

Now $x' \mid N$, hence $\gcd(x', k) = 1$, and $\gcd(x', z') = 1$. Thus the last divisibility is equivalent to $x' \mid M$, which is exactly (5.1). $\square$

Corollary 5.2. *For every $r \geq 1$,*

$$\mathcal{W}_r \subseteq \mathcal{W}_{r+1}.$$

More precisely, if a width occurs for a squarefree integer with r prime factors, then it occurs for infinitely many squarefree integers with $r+1$ prime factors.

Proof. Apply Lemma 5.1 with $M = N$. Then every $x, z \in S(n)$ satisfy

$$x \mid (q+1)z,$$

so

$$S(nq) = S(n) \oplus (q+1)S(n)$$

is the ordinal sum of two copies of $S(n)$. Its width is therefore $a(n)$. The lemma gives infinitely many choices of q . $\square$

To realize the missing widths between b_r and b_{r+1} , start from an exact Boolean lattice and add one prime whose cross-relations between the two Boolean copies are controlled by a single integer threshold.

Fix $r \geq 1$ and put

$$A = 2^r, \quad w_i = A + 2^{i-1} \quad (1 \leq i \leq r).$$

Choose distinct auxiliary primes $\ell_1, \dots, \ell_r > 3$. By the Chinese remainder theorem and Dirichlet's theorem, we may choose distinct primes $p_1, \dots, p_r > 3$ such that

$$p_i \equiv 3^{w_i} - 1 \pmod{3^{w_i+1}}, \tag{5.2}$$

$$p_i \equiv -1 \pmod{\ell_i}, \quad p_i \equiv 1 \pmod{\ell_j} \quad (j \neq i). \tag{5.3}$$

The residue classes are reduced, so infinitely many choices are available. Put

$$n = p_1 \cdots p_r, \quad x_I = \prod_{i \in I} (p_i + 1) \quad (I \subseteq [r]).$$

Equation (5.3) makes ℓ_i a private prime divisor of $p_i + 1$. Consequently

$$x_I \mid x_J \iff I \subseteq J, \tag{5.4}$$

so $S(n)$ is exactly a Boolean lattice B_r . By (5.2),

$$v_3(p_i + 1) = w_i.$$

Define

$$W(I) = v_3(x_I) = A|I| + \sum_{i \in I} 2^{i-1}. \tag{5.5}$$

These weights are pairwise distinct integers. Indeed, for fixed $|I|$ the binary sum $\sum_{i \in I} 2^{i-1}$ determines I , while different cardinalities lie in disjoint intervals $[kA, (k+1)A - 1]$. Let

$$H = W([r]) = v_3(\sigma(n)).$$

For $0 \leq t \leq H$, put

$$M_t = \frac{\sigma(n)}{3^{H-t}}.$$

Since the primes p_i are odd, M_t is even. Lemma 5.1 therefore gives infinitely many primes $q_t > \sigma(n)$ satisfying its conclusion with $M = M_t$.

Let $P_t = S(nq_t)$ as a divisibility poset. It consists of lower and upper copies x_I and $(q_t + 1)x_J$ of B_r ; since $q_t > \sigma(n)$ there are no divisibilities from the upper copy to the lower. By (5.1), only the 3-adic part can obstruct a lower-to-upper divisibility, and

$$v_3\left(\frac{x_I}{\gcd(x_I, x_J)}\right) = \max\{W(I) - W(J), 0\}.$$

Hence the cross-relations are exactly

$$\boxed{x_I \mid (q_t + 1)x_J \iff W(I) - W(J) \leq t.} \quad (5.6)$$

Lemma 5.3 (Width interpolation). *Let $\alpha_t = \text{width}(P_t)$. Then*

$$\alpha_{t+1} \leq \alpha_t \leq \alpha_{t+1} + 1 \quad (0 \leq t < H). \quad (5.7)$$

Moreover,

$$\alpha_0 = b_{r+1}, \quad \alpha_H = b_r. \quad (5.8)$$

Consequently every integer between b_r and b_{r+1} occurs among the values α_t .

Proof. For an integer s , let

$$I_s = \{I \subseteq [r] : W(I) \leq s\}, \quad F_s = \{I \subseteq [r] : W(I) > s\},$$

with the induced Boolean order. We claim that

$$\alpha_t = \max_{s \in \mathbb{Z}} (\text{width}(I_s) + \text{width}(F_{s+t})). \quad (5.9)$$

Indeed, split an antichain of P_t into its upper and lower parts. If the upper part is nonempty, choose an upper element x_J of largest weight and put $s = W(J)$. Every upper element then belongs to I_s . If a lower element x_I had $W(I) \leq s + t$, then (5.6) would give $x_I \mid (q_t + 1)x_J$, contradicting the antichain property. Hence every lower element belongs to F_{s+t} . The two parts therefore have sizes at most $\text{width}(I_s)$ and $\text{width}(F_{s+t})$. If the upper part is empty, take $s < -t$, so that I_s is empty and F_{s+t} is the whole Boolean lattice. If the lower part is empty, take $s \geq H$, so that I_s is the whole Boolean lattice and F_{s+t} is empty. Conversely, choose maximal antichains $A \subseteq I_s$ and $B \subseteq F_{s+t}$, place A in the upper copy and B in the lower copy, and consider $I \in B$, $J \in A$. Then $W(I) > s + t \geq W(J) + t$, so (5.6) excludes every lower-to-upper comparability. Thus $A \cup B$ is an antichain. This proves (5.9).

The weights (5.5) are pairwise distinct integers. Therefore, for fixed s , passing from F_{s+t} to F_{s+t+1} deletes at most one element. Deleting one element from a finite poset can lower its width by at most one. Formula (5.9) then gives

$$\alpha_t - 1 \leq \alpha_{t+1} \leq \alpha_t,$$

which is (5.7).

For $t = H$, condition (5.6) holds for every pair I, J , so every lower element lies below every upper element. Thus P_H is the ordinal sum of two copies of B_r and

$$\alpha_H = b_r.$$

For $t = 0$, identify the lower copy with the layer of B_{r+1} having last coordinate 0, and the upper copy with the layer having last coordinate 1. Comparabilities within each layer are the Boolean ones. A cross-comparability in B_{r+1} has the form $(I, 0) \leq (J, 1)$ with $I \subseteq J$; then $W(I) \leq W(J)$, so (5.6) gives the same comparability in P_0 . Thus every comparability of B_{r+1} occurs in P_0 , and hence

$$\alpha_0 \leq b_{r+1}.$$

Put

$$h = \left\lfloor \frac{r+1}{2} \right\rfloor.$$

Every h -element subset has weight at least hA , whereas every $(h-1)$ -element subset has weight at most

$$(h-1)A + (A-1) = hA - 1.$$

Therefore the h -th rank of the lower copy together with the $(h-1)$ -st rank of the upper copy is an antichain in P_0 . Its size is

$$\binom{r}{h} + \binom{r}{h-1} = \binom{r+1}{h} = b_{r+1}.$$

Thus $\alpha_0 = b_{r+1}$. Since the integer sequence α_t decreases from b_{r+1} to b_r in steps of size at most one, it assumes every intermediate integer value. $\square$

Theorem 5.4 (Complete squarefree spectrum). *For every $r \geq 1$,*

$$\boxed{\mathcal{W}_r = \left\{ 1, 2, \dots, \binom{r}{\lfloor r/2 \rfloor} \right\}}. \quad (5.10)$$

Moreover, every value in this interval is attained by infinitely many squarefree integers with exactly r prime factors.

Proof. We argue by induction on r . The assertion is immediate for $r = 1$. Assume it for r . Corollary 5.2 preserves every value $1, \dots, b_r$ when one new prime factor is added, with infinitely many realizations. Lemma 5.3 and the construction preceding it realize every value

$$b_r, b_r + 1, \dots, b_{r+1}$$

with $r+1$ prime factors, again with infinitely many realizations because the final prime q_t can be chosen in infinitely many ways. Thus every value from 1 to b_{r+1} occurs. The upper bound in Theorem 3.1 shows that no larger value is possible. $\square$

6 Optimal representations of divisibility posets

By Theorem 4.1, repeated exponent blocks lead to widths of powers P^r . Engel's limit theory expresses their asymptotics through the minimum representation variance $V(P)$; we therefore determine $V(D_m)$.

For a finite poset P with uniform weight, a *representation* is a function

$$x : P \rightarrow \mathbb{R}$$

such that

$$x(q) - x(p) \geq 1 \quad (p < q).$$

Let $V(P)$ be the minimum of the uniform variance $\text{Var}(x)$ over all representations x , and put

$$V_m = V(D_m).$$

Engel's asymptotic product theorem states that for every finite poset P which is not an antichain,

$$\text{width}(P^r) \sim \frac{|P|^r}{\sqrt{2\pi r V(P)}} \quad (r \rightarrow \infty)$$

under uniform weights [4, 6]; see in particular [6, Theorem 7.2.1]. Our task is therefore to determine $V(D_m)$.

6.1 A Ferrers lemma

For graded posets, filter-average conditions are closely related to rank compression; see Bezrukov and Engel [1]. Since our Ferrers truncations need not be graded, we prove directly the filter-average inequality needed for optimality.

For integers

$$w_0 > w_1 > \cdots > w_{H-1} > 0, \quad w_H = 0,$$

let

$$P(w) = \{(a, b) : 0 \leq b < H, 0 \leq a < w_b\}$$

with the coordinatewise order, and define the natural height

$$\rho(a, b) = a + b.$$

Lemma 6.1 (Ferrers lemma). *Assume*

$$w_b - w_{b+1} \in \{1, 2\} \quad (0 \leq b < H).$$

Then ρ is an optimal representation of $P(w)$.

It suffices to show that every filter has average height at least the global mean; then every monotone perturbation of ρ has nonnegative covariance with ρ .

Proof. Let μ be the average of ρ on $P(w)$. We first prove that every order ideal has average height at most μ .

An ideal is described by row lengths

$$w_b \geq x_b \geq x_{b+1} \geq 0, \quad x_H := 0.$$

Relax $x_0, \dots, x_{H-1}$ to real variables and let K be the resulting compact polytope. Define

$$\Phi(x) = \sum_{b=0}^{H-1} \left(bx_b + \frac{x_b(x_b - 1)}{2} - \mu x_b \right).$$

For integral x , this is the total value of $\rho - \mu$ on the corresponding ideal. The function Φ is convex, so it is enough to prove $\Phi \leq 0$ at the vertices of K .

At a vertex, every maximal constant block

$$x_s = \dots = x_t = c$$

is pinned either at $c = 0$ or at $c = w_t$. Indeed, if $0 < c < w_t$, maximality leaves positive slack in the neighboring monotonicity constraints (with the evident boundary modifications), and the upper bounds are also slack because $w_b \geq w_t > c$ on the block. For sufficiently small $\varepsilon > 0$, shifting the whole block by $\pm \varepsilon$ gives two feasible points with midpoint x , contradicting extremality.

A positive constant block $x_s = \dots = x_t = w_t$ contributes to the complement precisely the cells in rows $s, \dots, t-1$ lying to the right of column $w_t - 1$; a final zero block contributes the remaining full rows. Thus the complement of a vertex ideal is a disjoint union of wedges

$$W_{s,t} = \{(a, b) : s \leq b < t, w_t \leq a < w_b\}, \quad 0 \leq s < t \leq H.$$

We show that every such wedge has average height at least μ .

Fix s . Inside $W_{s,t}$, the average height of row b is

$$b + \frac{w_t + w_b - 1}{2}, \quad s \leq b < t.$$

These row averages are nondecreasing in b , because $w_b - w_{b+1} \leq 2$. For $t < H$,

$$W_{s,t+1} = W_{s,t} \dot{\cup} R_{s,t},$$

where

$$R_{s,t} = \{(a, b) : w_{t+1} \leq a < w_t, s \leq b \leq t\}.$$

The smallest row average in $W_{s,t}$ exceeds the average height of $R_{s,t}$ by

$$\frac{s + w_s - w_{t+1} - t}{2} \geq \frac{1}{2},$$

because every row-length drop is at least 1, so $w_s - w_{t+1} \geq t - s + 1$. Hence adjoining $R_{s,t}$ can only decrease the average, and therefore

$$\text{av}(W_{s,t}) \geq \text{av}(W_{s,H}).$$

Now $W_{s,H}$ is simply the union of the full rows $s, \dots, H-1$. The average height of the full b -th row is

$$b + \frac{w_b - 1}{2},$$

and these row averages are again nondecreasing. Removing the initial rows can therefore only increase the overall average, so

$$\text{av}(W_{s,H}) \geq \mu.$$

Thus every wedge has average height at least μ . The complement of every vertex ideal has average at least μ , hence every vertex ideal has average at most μ . Since Φ is convex, the same holds for every ideal.

Equivalently, every nonempty filter has average ρ at least μ . Let x be any representation and write

$$x = \rho + y.$$

The set $P(w)$ is an order ideal in $\mathbb{N}_0^2$, so every cover changes exactly one coordinate by 1. Hence ρ increases by exactly 1 along every cover relation, and y is nondecreasing. Every nondecreasing function on a finite poset is a constant plus a nonnegative linear combination of indicator functions of filters: list its distinct values $c_0 < \cdots < c_m$ and write

$$y = c_0 + \sum_{j=1}^m (c_j - c_{j-1}) \mathbf{1}_{\{y \geq c_j\}},$$

where each upper level set $\{y \geq c_j\}$ is a filter. It follows from the filter inequality that

$$\text{Cov}(\rho, y) \geq 0.$$

Therefore

$$\text{Var}(x) = \text{Var}(\rho) + 2 \text{Cov}(\rho, y) + \text{Var}(y) \geq \text{Var}(\rho).$$

Thus ρ is optimal. □

6.2 The logarithmic Ferrers diagrams

For $T \geq 1$, define

$$Q_T = \{(a, b) \in \mathbb{N}_0^2 : 2^a 3^b \leq T\}.$$

Put

$$N(T) = |Q_T|, \quad \mu(T) = \frac{1}{N(T)} \sum_{(a,b) \in Q_T} (a+b),$$

and

$$v(T) = \frac{1}{N(T)} \sum_{(a,b) \in Q_T} (a+b - \mu(T))^2.$$

Figure 2 shows the shape to keep in mind.

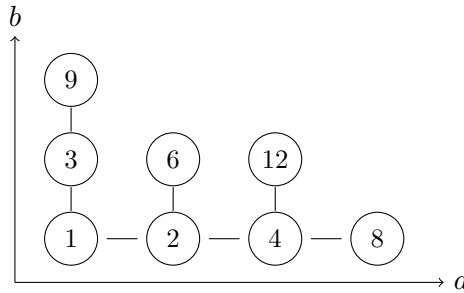


Figure 2: The diagram Q_{12} . The label at (a, b) is $2^a 3^b$. Moving right multiplies by 2, moving up multiplies by 3, and the natural height is $a + b$. The row lengths are 4, 3, 1.

The nonempty rows of Q_T have lengths

$$w_b = 1 + \left\lfloor \log_2 \frac{T}{3^b} \right\rfloor.$$

Since $1 < \log_2 3 < 2$, successive nonempty row lengths differ by 1 or 2; the last nonempty row has length 1 or 2. Lemma 6.1 therefore applies to every Q_T .

We need one quantitative consequence.

Lemma 6.2 (Scaling lemma). *For every integer $T \geq 1$,*

$$\mu(2T) - \mu(T) \geq \frac{1}{2}.$$

Consequently,

$$\mu(4T) - \mu(T) \geq 1.$$

Proof. Let h be the number of points of the first column of Q_{2T} , equivalently

$$h = \#\{b \geq 0 : 3^b \leq 2T\}.$$

For $0 \leq b < h$, let w_b be the number of $a \geq 0$ such that $2^a 3^b \leq T$; the last w_b is allowed to be 0. Put

$$N = N(T) = \sum_{b=0}^{h-1} w_b$$

and

$$c_b = w_b + 2b.$$

The sequence (w_b) is nonincreasing, while

$$c_{b+1} - c_b = 2 - (w_b - w_{b+1}) \in \{0, 1\}$$

whenever both sides are defined. Hence (c_b) is nondecreasing. Since (w_b) and (c_b) are oppositely ordered,

$$h \sum_b w_b c_b - N \sum_b c_b = \sum_{0 \leq i < j < h} (w_i - w_j)(c_i - c_j) \leq 0.$$

Therefore

$$\frac{\sum_b w_b c_b}{N} \leq \frac{1}{h} \sum_b c_b.$$

Now

$$2\mu(T) + 1 = \frac{\sum_b w_b c_b}{N},$$

whereas

$$\sum_b c_b = N + h(h-1).$$

Therefore

$$\mu(T) \leq \frac{1}{2} \left(\frac{N}{h} + h - 2 \right). \quad (6.1)$$

The set Q_{2T} is the disjoint union of the translate

$$(a, b) \mapsto (a+1, b)$$

of Q_T and the first column

$$\{(0, b) : 0 \leq b < h\}.$$

Thus

$$N(2T) = N + h$$

and

$$\mu(2T) - \mu(T) = \frac{N + h((h-1)/2 - \mu(T))}{N + h}.$$

By (6.1),

$$\frac{h-1}{2} - \mu(T) \geq \frac{h-N}{2h}.$$

The numerator above is therefore at least

$$N + \frac{h-N}{2} = \frac{N+h}{2},$$

which proves

$$\mu(2T) - \mu(T) \geq \frac{1}{2}.$$

Applying this twice gives the second assertion. $\square$

Lemma 6.3. *The function $T \mapsto \mu(T)$ is nondecreasing on the positive integers.*

Proof. If $S \leq T$, then Q_S is an order ideal of Q_T . By Lemma 6.1, every ideal of Q_T has average height at most the average height of Q_T . Hence $\mu(S) \leq \mu(T)$. $\square$

6.3 Critical dilation

The scaling lemma already gives the precise dilation needed for the prime-core argument.

Corollary 6.4 (Critical dilation). *The least positive integer L such that*

$$\mu(LT) - \mu(T) \geq 1 \quad (T \geq 1)$$

is

$$\boxed{L = 4.}$$

Proof. Lemma 6.2, applied twice, gives

$$\mu(4T) - \mu(T) \geq 1$$

for every T . To see that no smaller integer works, take $T = 1$. For $L = 1, 2, 3$, the sets Q_L have mean heights 0, $1/2$, and $2/3$, respectively, all strictly below $1 = \mu(1) + 1$. $\square$

Thus the unit-gain scale for the 2,3-smooth truncations is exactly 4. Since the next prime is 5, this is the numerical separation that makes all prime directions at least 5 redundant in the optimal representation.

6.4 Prime-core reduction

For a finite set $\mathcal{P}$ of primes, let $D_m^{(\mathcal{P})}$ be the poset on $\{1, \dots, m\}$ in which $a \preceq b$ when $a \mid b$ and every prime divisor of b/a belongs to $\mathcal{P}$. Also put

$$S_{\mathcal{P}}(T) = \{n \leq T : \text{every prime divisor of } n \text{ belongs to } \mathcal{P}\},$$

ordered by divisibility, let $\rho(n) = \Omega(n)$, and write $\mu_{\mathcal{P}}(T)$ for the average of ρ on $S_{\mathcal{P}}(T)$.

The mechanism behind the 2,3-reduction is the following general principle.

Theorem 6.5 (Prime-core reduction principle). *Suppose that*

- (i) ρ is an optimal representation of $S_{\mathcal{P}}(T)$ for every $T \geq 1$;

(ii) $\mu_{\mathcal{P}}(T)$ is nondecreasing;

(iii) for some integer $L \geq 2$,

$$\mu_{\mathcal{P}}(LT) - \mu_{\mathcal{P}}(T) \geq 1 \quad (T \geq 1);$$

(iv) every prime outside $\mathcal{P}$ is at least L .

Then

$$V(D_m) = V(D_m^{(\mathcal{P})}) \quad (m \geq 1).$$

More precisely, write uniquely $n = us$, where s is $\mathcal{P}$ -smooth and u has no prime divisor in $\mathcal{P}$.

Then

$$x_m(n) = \Omega(s) - \mu_{\mathcal{P}}\left(\left\lfloor \frac{m}{u} \right\rfloor\right)$$

is an optimal representation of D_m .

Proof. For fixed u , the corresponding component of $D_m^{(\mathcal{P})}$ is isomorphic to

$$S_{\mathcal{P}}\left(\left\lfloor \frac{m}{u} \right\rfloor\right).$$

By (i), ρ is optimal on each component. Since different components are disconnected, they may be translated independently. Centering each component at mean 0 therefore gives the displayed optimal representation of $D_m^{(\mathcal{P})}$.

It remains to check the missing divisibility relations. Suppose $n < n'$ and $n \mid n'$. Write

$$n = us, \quad n' = uvs w,$$

where s, w are $\mathcal{P}$ -smooth and u, v have no prime divisor in $\mathcal{P}$. Put

$$T = \left\lfloor \frac{m}{u} \right\rfloor, \quad T' = \left\lfloor \frac{m}{uv} \right\rfloor.$$

Then

$$x_m(n') - x_m(n) = \Omega(w) + \mu_{\mathcal{P}}(T) - \mu_{\mathcal{P}}(T').$$

If $w > 1$, then $\Omega(w) \geq 1$, and (ii) gives a difference at least 1.

If $w = 1$, then $v > 1$. By (iv), $v \geq L$, so $LT' \leq T$. Hence (ii) and (iii) give

$$\mu_{\mathcal{P}}(T) - \mu_{\mathcal{P}}(T') \geq \mu_{\mathcal{P}}(LT') - \mu_{\mathcal{P}}(T') \geq 1.$$

Thus the centered representation is feasible for the full poset D_m . Since adding constraints can only increase the minimum variance, it is optimal for both posets. $\square$

We now apply the principle to $\mathcal{P} = \{2, 3\}$. In this case $S_{\mathcal{P}}(T)$ is identified with Q_T , so optimality follows from Lemma 6.1, monotonicity from Lemma 6.3, and the critical dilation is 4 by Corollary 6.4. The first omitted prime is 5.

Theorem 6.6 (2, 3-reduction). *For every $m \geq 1$,*

$$\boxed{V(D_m) = V(D_m^{(2,3)})}.$$

More precisely, if

$$n = u2^a 3^b, \quad (u, 6) = 1,$$

then

$$\boxed{x_m(n) = a + b - \mu\left(\left\lfloor \frac{m}{u} \right\rfloor\right)} \tag{6.2}$$

is an optimal representation of D_m .

Proof. Apply Theorem 6.5 with $\mathcal{P} = \{2, 3\}$ and $L = 4$. □

The arithmetic reason is therefore concentrated in

$$\boxed{3 < 4 < 5.}$$

The prime 3 lies below the unit-gain scale $4 = 2^2$, while the next prime already lies beyond it. Thus the 2- and 3-directions must be retained, but every prime direction at least 5 is absorbed by the centering distance between components.

The pair $\{2, 3\}$ is sharp in two complementary senses.

Proposition 6.7 (Minimal universal prime core). *For a set $\mathcal{P}$ of primes,*

$$\boxed{V(D_m^{(\mathcal{P})}) = V(D_m) \text{ for every } m \iff \{2, 3\} \subseteq \mathcal{P}.} \quad (6.3)$$

Proof. If $\{2, 3\} \subseteq \mathcal{P}$, then

$$D_m^{(2,3)} \subseteq D_m^{(\mathcal{P})} \subseteq D_m$$

in the sense of inclusion of order relations. Adding constraints can only increase the minimum representation variance, so Theorem 6.6 gives equality throughout.

Conversely, if $2 \notin \mathcal{P}$, then $D_2^{(\mathcal{P})}$ is an antichain and has optimal variance 0, whereas D_2 is a two-element chain and has optimal variance $1/4$. Suppose next that $2 \in \mathcal{P}$ but $3 \notin \mathcal{P}$. Then $D_3^{(\mathcal{P})}$ consists of the chain $1 < 2$ and the isolated point 3, so its optimal variance is $1/6$. For the full poset D_3 , the centered optimal representation has values $-2/3, 1/3, 1/3$, and hence

$$V(D_3) = \frac{2}{9} > \frac{1}{6}.$$

Thus both 2 and 3 are necessary. □

The proposition shows that $\{2, 3\}$ is the smallest universal set of prime directions that determines the exact variance of every D_m .

6.5 Exact formula and limiting constant

Theorem 6.6 immediately gives an exact finite formula.

Corollary 6.8. *For every $m \geq 1$,*

$$\boxed{V_m = \frac{1}{m} \sum_{\substack{u \leq m \\ (u,6)=1}} N\left(\left\lfloor \frac{m}{u} \right\rfloor\right) v\left(\left\lfloor \frac{m}{u} \right\rfloor\right).} \quad (6.4)$$

Proof. Each component indexed by u has $N(\lfloor m/u \rfloor)$ elements and, after centering, variance $v(\lfloor m/u \rfloor)$. All component means are 0, so the variance of the union is the size-weighted average of the component variances. □

Theorem 6.9. *The sequence V_m converges. More precisely,*

$$\boxed{V_m = C + O\left(\frac{(\log(2m))^4}{m}\right),}$$

where

$$C = \frac{1}{3} \sum_{t \geq 1} \frac{N(t)v(t)}{t(t+1)}. \quad (6.5)$$

Numerically,

$$C = 1.0367934201598 \dots$$

Proof. Put

$$F(t) = N(t)v(t).$$

Since both coordinates in Q_t are $O(\log(2t))$,

$$F(t) = O((\log(2t))^4),$$

so the series in (7.10) converges absolutely.

We first note that $F(t)$ is nondecreasing. When a new point of height h is added to a set of N heights with mean μ and unnormalized variance F , the new unnormalized variance is

$$F + \frac{N}{N+1}(h - \mu)^2.$$

Thus enlarging Q_t cannot decrease $F(t)$.

Let

$$B(x) = \#\{u \leq x : (u, 6) = 1\}.$$

Periodicity modulo 6 gives

$$B(x) = \frac{x}{3} + E(x),$$

where E is bounded on $[0, \infty)$. Grouping Corollary 6.8 according to

$$t = \left\lfloor \frac{m}{u} \right\rfloor$$

gives

$$V_m = \frac{1}{m} \sum_{t=1}^m F(t) A_m(t),$$

where

$$A_m(t) = B(m/t) - B(m/(t+1)).$$

Hence

$$A_m(t) = \frac{m}{3t(t+1)} + E(m/t) - E(m/(t+1)).$$

Write $e_t = E(m/t)$. The discrepancy from the first m terms of (7.10) is therefore

$$\frac{1}{m} \sum_{t=1}^m F(t)(e_t - e_{t+1}).$$

Summation by parts yields

$$\sum_{t=1}^m F(t)(e_t - e_{t+1}) = F(1)e_1 + \sum_{t=2}^m (F(t) - F(t-1))e_t - F(m)e_{m+1}.$$

Since E is bounded and F is nondecreasing, the right-hand side is $O(F(m))$. Thus the discrepancy is

$$O\left(\frac{(\log(2m))^4}{m}\right).$$

Finally,

$$\sum_{t>m} \frac{F(t)}{t(t+1)} = O\left(\frac{(\log(2m))^4}{m}\right)$$

by comparison with the corresponding integral. Combining the two estimates proves the theorem. $\square$

Remark 6.10. The constant C has a probabilistic interpretation. Let $\mathcal{H} = \{2^a 3^b : a, b \geq 0\}$, choose $N \in \mathcal{H}$ with

$$\Pr(N = n) = \frac{1}{3n},$$

let U be independent and uniform on $(0, 1)$, and put $T = \lfloor N/U \rfloor$. For $n \leq t$,

$$\Pr(N = n, T = t) = \frac{1}{3t(t+1)},$$

so conditional on $T = t$, the variable N is uniform on the 3-smooth integers at most t . Hence

$$C = \mathbb{E}[\text{Var}(\Omega(N) \mid T)].$$

Writing $N = 2^A 3^B$ shows that A and B are independent geometric variables with $\text{Var}(A+B) = 11/4$. The law of total variance therefore gives

$$\boxed{\frac{11}{4} = C + \text{Var}(\mu(T)).}$$

Thus C measures the residual variance of $\Omega(N)$ after the truncation parameter T is known.

6.6 Repeated exponent patterns

The preceding variance calculation applies to more than equal exponents. Fix an exponent vector

$$\mathbf{e} = (e_1, \dots, e_s), \quad e_j \geq 1,$$

and let $M_{\mathbf{e}}(r)$ be the largest value of $a(n)$ among integers whose prime-exponent multiset consists of r copies of $\mathbf{e}$. Thus

$$n = \prod_{\nu=1}^r \prod_{j=1}^s p_{\nu,j}^{e_j},$$

with all primes $p_{\nu,j}$ distinct. Put

$$P_{\mathbf{e}} = D_{e_1+1} \times \cdots \times D_{e_s+1}.$$

By Theorem 4.1,

$$M_{\mathbf{e}}(r) = \text{width}(P_{\mathbf{e}}^r).$$

Thus a repeated exponent pattern is the ordinary power of one fixed finite poset, so Engel's asymptotic product theorem applies directly. His product theorem for optimal representations gives

$$V(P \times Q) = V(P) + V(Q)$$

for finite uniformly weighted posets [6, Theorem 4.6.6]. Hence

$$V(P_{\mathbf{e}}) = \sum_{j=1}^s V_{e_j+1}.$$

We obtain the following.

Corollary 6.11. *For every fixed exponent vector $\mathbf{e} = (e_1, \dots, e_s)$,*

$$M_{\mathbf{e}}(r) \sim \frac{\left(\prod_{j=1}^s (e_j + 1)\right)^r}{\sqrt{2\pi r \sum_{j=1}^s V_{e_j+1}}} \quad (r \rightarrow \infty).$$

Proof. By Theorem 4.1, $M_{\mathbf{e}}(r) = \text{width}(P_{\mathbf{e}}^r)$. Engel's asymptotic product theorem therefore gives

$$M_{\mathbf{e}}(r) \sim \frac{|P_{\mathbf{e}}|^r}{\sqrt{2\pi r V(P_{\mathbf{e}})}}.$$

Since $|P_{\mathbf{e}}| = \prod_{j=1}^s (e_j + 1)$ and $V(P_{\mathbf{e}}) = \sum_{j=1}^s V_{e_j+1}$, the stated formula follows. $\square$

7 Minimum widths and open problems

The maximal problem is governed by eliminating cross-divisibilities; the opposite extremal question asks how strongly they can collapse the universal product poset. For an exponent vector $\mathbf{e} = (e_1, \dots, e_r)$, put

$$L(\mathbf{e}) = \min_{p_1, \dots, p_r} a(p_1^{e_1} \cdots p_r^{e_r}),$$

where the minimum is over distinct primes.

For a single exponent, Theorem 2.3 gives

$$L(e) = \left\lceil \frac{e+1}{2} \right\rceil.$$

A simple construction gives

$$L(e, 2) \leq 2L(e).$$

Indeed, for $m = p^e$ let M be the least common multiple of the elements of $S(m)$ and choose a prime $q \equiv -1 \pmod{M}$. Then

$$S(mq^2) = S(m) \cup (q+1)S(m) \cup (q^2+q+1)S(m),$$

and a minimum chain cover of $S(m)$ covers the first two copies consecutively, while a second copy of the same cover handles the third. Exact computations in the first cases suggest that this upper bound is sharp.

Conjecture 7.1. *For every $e \geq 1$,*

$$L(e, 2) = 2 \left\lceil \frac{e+1}{2} \right\rceil.$$

More generally, determining $L(e_1, \dots, e_r)$ asks how far arithmetic cross-divisibilities can collapse the universal index poset. This completes the contrast running through the paper: in the maximal problem cross-divisibilities are eliminated, in the squarefree spectrum they are tuned through a threshold, and in the variance problem only the prime directions 2 and 3 remain active.

Use of artificial intelligence

OpenAI ChatGPT was used to suggest possible proof approaches, stress-test selected arguments, and assist with organization and wording. The author independently verified the mathematical claims, computations, and references and takes full responsibility for the final content.

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